

The Pratyaksh Framework:

A Self-Limiting TVD Explicit Runge-Kutta Family for
Real-Time Physics, Robotics, and Scientific Computing

Pratyaksh Raj

pratyakshnarayanlal1@gmail.com

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Abstract

Explicit Runge-Kutta methods, most notably the classical fourth-order scheme (RK4), remain the computational foundation of dynamical simulation across robotics, computer graphics, and machine learning due to their high accuracy, matrix-free evaluation, and branchless memory layout. However, explicit polynomial integrators suffer from catastrophic numerical divergence (NaN overflow) whenever localized physical transients, contact impulses, or steep shock gradients violate linear stability boundaries ($|z| > 2.785$). Conversely, unconditionally stable implicit integrators (such as Radau IIA or BDF) require constructing and inverting $N \times N$ Jacobian matrices, rendering them computationally prohibitive for high-dimensional spatial discretizations ($N > 10^4$), unpredictable for hard real-time deadlines (60/120 Hz), and ill-suited for massively parallel SIMD GPU architectures.

In this paper, we introduce **Pratyaksh**, an explicit, matrix-free rational Runge-Kutta framework that dynamically bounds stage divergence through a dimensionless vector curvature ratio. By evaluating the denominator via a global inner-product norm, we resolve the classical coordinate-singularity pathology that historically plagued rational integrators at physical turning points ($\vec{K}_1 \rightarrow 0$). We derive two canonical formulations: **Formula A** (Order 2), which introduces $O(\Delta t^2)$ non-linear viscosity providing strict Total Variation Diminishing (TVD) shock-capturing (0 TV increases across 500 shock steps, 0.44% peak error on a 2,000-dimensional Brusselator wave), and **Formula B** (Order 4), which cancels intermediate Taylor terms to preserve asymptotic fourth-order convergence ($p = 4.000$) down to machine precision (10^{-12}).

Through discrete Fourier dispersion analysis of the biharmonic Kuramoto-Sivashinsky operator, we demonstrate that the linear stability boundary matches classical RK4 ($z = -2.78529$) to four decimal places ($\Delta t_{\text{crit}} = 0.004306$ s). We characterize the extreme under-resolution regime ($|z| \gg 2.79$) as a sharp knife-edge bifurcation where floating-point overflow is converted into an arrested time-freeze, and formulate a zero-overhead concurrent diagnostic ($\text{den} > 5.0$) that alerts supervisory control loops without requiring out-of-band stiffness estimation. The Pratyaksh framework bridges the long-standing gap between fragile explicit polynomials and unscalable implicit systems for fixed-step computational engineering.

1 Introduction

Numerical integration of initial value problems (IVPs) governed by systems of ordinary differential equations (ODEs),

$$\frac{d\vec{y}}{dt} = \vec{f}(t, \vec{y}), \quad \vec{y}(t_0) = \vec{y}_0 \in \mathbb{R}^d, \quad (1)$$

and semi-discrete partial differential equations (PDEs) is a cornerstone of scientific computing, aerospace engineering, real-time robotics, and physics-informed machine learning.

For over a century, the classical four-stage fourth-order explicit Runge-Kutta method (RK4) [1, 2] has stood as the default numerical engine for non-stiff systems. Its popularity stems from

its explicit, matrix-free structure: advances are computed strictly via forward evaluations of $\vec{f}(t, \vec{y})$ without requiring Jacobian matrices, linear algebra factorizations, or Newton-Raphson iteration loops.

However, explicit polynomial schemes possess a fatal mathematical vulnerability: their linear stability domain is strictly bounded. For the scalar Dahlquist test problem $y' = \lambda y$ [3], the stability polynomial $R(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}$ satisfies $|R(z)| \leq 1$ only within the real interval $z = \lambda \Delta t \in [-2.78529, 0]$. If local stiffness, steep spatial gradients, or sudden contact impulses push $|z|$ past this threshold, the polynomial diverges exponentially, producing catastrophic floating-point overflow ($\pm\infty$, NaN) that corrupts computational memory and crashes simulation pipelines.

In classical numerical analysis, this limitation is bypassed using **implicit methods** (e.g., Radau IIA, Gauss-Legendre, or backward differentiation formulas) [8]. Implicit methods achieve unconditional A -stability by solving algebraic systems of the form:

$$\vec{y}_{n+1} = \vec{y}_n + \Delta t \sum_{i=1}^s b_i \vec{f}(t_n + c_i \Delta t, \vec{Y}_i). \quad (2)$$

While mathematically unyielding, implicit methods fail to meet the operational constraints of three major pillars of modern computational science:

1. **Hard Real-Time Robotics and Physics Engines (60/120 Hz):** In game physics (PhysX, Unreal Engine) and robotics simulation (MuJoCo, Isaac Sim), each simulation step must finish within a hard deadline of 8.33 to 16.67 ms. Implicit solvers rely on Newton iterations whose convergence rates vary unpredictably, causing severe frame drops.
2. **Massively Parallel SIMD GPUs (AI / Neural ODEs):** On modern GPU architectures (NVIDIA H100) running 10^5 parallel environments, warp efficiency requires branchless, uniform execution. Iterative linear solvers induce catastrophic thread divergence.
3. **High-Dimensional PDE Discretizations ($N > 10^5$):** Building, storing, and inverting an $N \times N$ Jacobian matrix requires enormous memory and floating-point throughput ($O(N^3)$ dense, $O(N^{1.5})$ sparse), completely exhausting RAM.

Attempts to construct explicit rational Runge-Kutta (RRK) methods date back to Wambecq [4] and Hairer [5, 6]. Hairer’s RAT scheme [5] and follow-on work on parabolic systems [6] established nonlinear stability results for explicit rational Runge-Kutta formulations without requiring Jacobian storage. However, subsequent analysis by Peirce and Prevost [7] demonstrated that unconditionally stable explicit RRK schemes of this class suffer from a fundamental lack of convergence under certain model problems, motivating renewed scrutiny of the accuracy-stability trade-off that the Pratyaksh framework’s dual-formula taxonomy (Section 3) directly addresses.

In this paper, we present **The Pratyaksh Framework**, an explicit, matrix-free, vector-norm rational Runge-Kutta architecture that guarantees positivity of its denominator, eliminates turning-point singularities, and provides a dual-formula taxonomy balancing TVD shock-capturing dissipation against asymptotic fourth-order precision.

2 The Pratyaksh Mathematical Framework

2.1 The Master Update Equation

Given state $\vec{y}_n \in \mathbb{R}^d$ and time-step Δt , the Pratyaksh framework evaluates the standard four Runge-Kutta stages:

$$\vec{K}_1 = \Delta t \vec{f}(t_n, \vec{y}_n), \quad (3)$$

$$\vec{K}_2 = \Delta t \vec{f}\left(t_n + \frac{1}{2}\Delta t, \vec{y}_n + \frac{1}{2}\vec{K}_1\right), \quad (4)$$

$$\vec{K}_3 = \Delta t \vec{f}\left(t_n + \frac{1}{2}\Delta t, \vec{y}_n + \frac{1}{2}\vec{K}_2\right), \quad (5)$$

$$\vec{K}_4 = \Delta t \vec{f}\left(t_n + \Delta t, \vec{y}_n + \vec{K}_3\right). \quad (6)$$

The state is advanced via the rational vector operator:

$$\vec{y}_{n+1} = \vec{y}_n + \frac{\frac{1}{6}(\vec{K}_1 + 2\vec{K}_2 + 2\vec{K}_3 + \vec{K}_4)}{1 + \frac{1}{2}\mathcal{D}(\vec{C}, \vec{K}_1)}, \quad (7)$$

where $\vec{C} \in \mathbb{R}^d$ is a discrete multi-stage curvature vector, and $\mathcal{D}(\vec{C}, \vec{K}_1)$ is a dimensionless discrepancy metric.

2.2 Coordinate Invariance via Vector Inner-Product Norm

In component-wise rational formulations, evaluating $\frac{C_i^2}{K_{1,i}^2 + \epsilon}$ causes catastrophic failure whenever an individual velocity component approaches zero ($K_{1,i} \rightarrow 0$), such as the apex of an oscillatory trajectory, artificially freezing individual coordinates.

The Pratyaksh framework resolves this by defining $\mathcal{D}$ using the global Euclidean inner-product norm:

$$\mathcal{D}(\vec{C}, \vec{K}_1) = \frac{\|\vec{C}\|^2}{\|\vec{K}_1\|^2 + \epsilon} = \frac{\sum_{i=1}^d C_i^2}{\sum_{i=1}^d K_{1,i}^2 + \epsilon}, \quad (8)$$

where $\epsilon = 10^{-14}$ is a machine-precision safeguard.

Theorem 1 (Pole-Free Guarantee). For all real vector fields $\vec{f} \in \mathbb{R}^d$, all step sizes $\Delta t > 0$, and all curvature vectors $\vec{C}$, the Pratyaksh denominator $\text{den} = 1 + \frac{1}{2}\mathcal{D}(\vec{C}, \vec{K}_1)$ satisfies:

$$\text{den} \geq 1.000000 \quad \forall \vec{y}_n \in \mathbb{R}^d. \quad (9)$$

Consequently, the Pratyaksh scheme possesses zero real poles and cannot divide by zero.

Proof. Because $\|\vec{C}\|^2 = \sum C_i^2 \geq 0$ and $\|\vec{K}_1\|^2 = \sum K_{1,i}^2 \geq 0$, the ratio $\mathcal{D} \geq 0$. Since $\beta = \frac{1}{2} > 0$, $\text{den} = 1 + \frac{1}{2}\mathcal{D} \geq 1$. $\square$

3 The Two Canonical Curvature Formulations

A fundamental law of numerical time-stepping is the **Order-Dissipation Trade-off**: an explicit rational scheme cannot simultaneously exhibit strong non-linear shock dissipation and asymptotic fourth-order convergence. The Pratyaksh framework addresses this via two specialized curvature operators:

3.1 Formula A: Order 2 TVD Dissipative / Shock-Capturing

$$\vec{C}_A = \vec{K}_4 - 2\vec{K}_3 + \vec{K}_2. \quad (10)$$

By Taylor expanding stages $\vec{K}_2, \vec{K}_3, \vec{K}_4$ around t_n for an autonomous scalar ODE $y' = f(y)$:

$$\begin{aligned} K_1 &= \Delta t f, \\ K_2 &= \Delta t f + \frac{1}{2}\Delta t^2 f f' + \frac{1}{8}\Delta t^3 f^2 f'' + O(\Delta t^4), \\ K_3 &= \Delta t f + \frac{1}{2}\Delta t^2 f f' + \frac{1}{8}\Delta t^3 f^2 f'' + \frac{1}{4}\Delta t^3 f(f')^2 + O(\Delta t^4), \\ K_4 &= \Delta t f + \Delta t^2 f f' + \frac{1}{2}\Delta t^3 f^2 f'' + \frac{1}{2}\Delta t^3 f(f')^2 + O(\Delta t^4). \end{aligned} \quad (11)$$

Substituting into $\vec{C}_A$:

$$C_A = K_4 - 2K_3 + K_2 = \frac{1}{2}\Delta t^2 f f' + \frac{3}{8}\Delta t^3 f^2 f'' + O(\Delta t^4). \quad (12)$$

Squaring C_A yields $C_A^2 = \frac{1}{4}\Delta t^4 f^2 (f')^2 + \frac{3}{8}\Delta t^5 f^3 f' f'' + O(\Delta t^6)$. Dividing by $K_1^2 = \Delta t^2 f^2$:

$$\text{den}_A = 1 + \frac{1}{2} \frac{C_A^2}{K_1^2} = 1 + \frac{1}{8}\Delta t^2 (f')^2 + \frac{3}{16}\Delta t^3 f f' f'' + O(\Delta t^4). \quad (13)$$

Expanding $(1 + \delta)^{-1} = 1 - \delta + O(\delta^2)$:

$$y_{n+1} = y_n + \Delta y_{\text{RK4}} \left(1 - \frac{1}{8}\Delta t^2 (f')^2 \right) + O(\Delta t^4). \quad (14)$$

Because the denominator injects an $O(\Delta t^2)$ perturbation into the update, the global error is bounded at $O(\Delta t^2)$ (**Order 2**).

However, this $O(\Delta t^2)$ term acts as a **non-linear artificial viscosity**, providing strict Total Variation Diminishing (TVD) shock dissipation [9].

3.2 Formula B: Order 4 Asymptotic High-Precision

$$\vec{C}_B = \vec{K}_4 - \vec{K}_3 - \vec{K}_2 + \vec{K}_1. \quad (15)$$

Examining the stage coefficients: $1 - 1 - 1 + 1 = 0$, both the $O(\Delta t)$ and $O(\Delta t^2)$ terms cancel identically:

$$C_B = \frac{1}{4}\Delta t^3 f [(f')^2 + f f''] + O(\Delta t^4). \quad (16)$$

Squaring C_B yields $C_B^2 = O(\Delta t^6)$. Dividing by $K_1^2 \sim O(\Delta t^2)$ gives:

$$\text{den}_B = 1 + \frac{1}{2} \frac{C_B^2}{K_1^2} = 1 + O(\Delta t^4). \quad (17)$$

Thus, the rational perturbation to the update is $O(\Delta t^5)$, preserving the local error at $O(\Delta t^5)$ and yielding a **pure fourth-order global convergence rate** ($p = 4.000$).

4 Stability and Bifurcation Analysis

4.1 Linear Stability Equivalence

Applying the Pratyaksh integrator to the linear scalar equation $y' = \lambda y$ with $z = \lambda \Delta t$:

$$K_1 = zy, \quad K_2 = zy(1 + \frac{1}{2}z), \quad K_3 = zy(1 + \frac{1}{2}z + \frac{1}{4}z^2), \quad K_4 = zy(1 + z + \frac{1}{2}z^2 + \frac{1}{4}z^3). \quad (18)$$

In the linear regime, the curvature $\vec{C}$ remains strictly proportional to $z^2 K_1$. Consequently, the real linear stability limit is governed by the classical condition $|R(z)| \leq 1$, yielding:

$$z_{\text{crit}} = -2.78529. \quad (19)$$

The Pratyaksh framework does not possess an artificially enlarged linear A -stability domain.

4.2 Knife-Edge Bifurcation and The Time-Freeze Mechanism

When local stiffness exceeds the linear threshold ($|z| > 2.785$), classical RK4 undergoes exponential divergence, producing NaN in fewer than 5 steps.

In the Pratyaksh framework, as stage divergence initiates, the ratio $\frac{\|\tilde{C}\|^2}{\|\tilde{K}_1\|^2}$ explodes to $10^{15} - 10^{24}$. The denominator divides the update by $\geq 10^{15}$, scaling the effective step size:

$$\Delta t_{\text{eff}} = \frac{\Delta t}{\text{den}} \rightarrow 0. \quad (20)$$

The Pratyaksh framework transforms a catastrophic floating-point crash (NaN) into a bounded, arrested time-freeze.

4.3 Zero-Cost Concurrent Runtime Diagnostic

Because den is already evaluated during the update step, it provides a zero-overhead **runtime stiffness monitor**:

$$\text{is_throttled} = (\text{den} > 5.0). \quad (21)$$

- $\text{den} \approx 1.0$: Simulation advances in unhindered physical time.
- $\text{den} > 5.0$: Downstream control logic is alerted that localized stiffness has triggered step throttling.

5 Numerical Benchmarks and Verification

5.1 Convergence Order Verification

We integrated the non-linear logistic equation $y' = 5y(1 - y)$ on $t \in [0, 2]$ across successive step-halving ($N = 50$ to $N = 3, 200$).

- **Formula B**: Achieved exact fourth-order convergence: $p = 4.000$, with errors scaling from 10^{-6} down to 1.2×10^{-12} .
- **Formula A**: Converged cleanly at second order: $p = 2.000$.

5.2 Kuramoto-Sivashinsky 4th-Order PDE: CFL Tracking

We discretized the chaotic biharmonic Kuramoto-Sivashinsky PDE:

$$u_t + uu_x + u_{xx} + u_{xxx} = 0, \quad x \in [0, 32\pi], \quad N = 256. \quad (22)$$

Discrete Fourier analysis yields the exact Nyquist dispersion relation for mode $k\Delta x = \pi$:

$$\lambda_{\min} = -D_2(\pi) - D_4(\pi) = -(-25.939) - (672.822) = -646.853 \text{ s}^{-1}. \quad (23)$$

The theoretical linear CFL boundary is:

$$\Delta t_{\text{crit}} = \frac{2.78529}{646.853} = \mathbf{0.004306 \text{ s}}. \quad (24)$$

Long-horizon simulations ($t = 20.0$ s, 4,600+ steps) confirm the empirical threshold:

- $\Delta t = 0.00430$ s: Stable ($\max |u| = 1.9319$)
- $\Delta t = 0.00431$ s: Stable ($\max |u| = 1.9319$)
- $\Delta t = 0.00432$ s: Classical RK4 explodes ($\max |u| = 50.40$). Formula A cushions the boundary.

5.3 Burgers' Shockwave: Strict TVD Verification

Testing viscous shock formation on $u_t + uu_x = \nu u_{xx}$ ($N = 500, \nu = 0.001$):

- Across 500 time steps through shock steepening, Formula A registered **0 Total Variation increases** (TV decayed monotonically from 4.000 to 2.897).
- Testing across three distinct initial waveforms (Single Sine, Bi-Harmonic, Gaussian packet) confirmed that non-linear shock steepening never triggers the denominator ($\text{den} \leq 1.0001$). The denominator activates exclusively when the linear diffusion limit $\Delta t_{\text{diff}} = \frac{2.785\Delta x^2}{4\nu}$ is exceeded.

5.4 2,000-Dimensional Brusselator Reaction-Diffusion PDE

At $\Delta t = 1.0 \times 10^{-4}$ s ($2.88\times$ beyond the classical RK4 explosion limit):

- Classical RK4 explodes to NaN in 9 steps.
- Formula A survives all 500 steps, capturing the wave peak with 0.44% **error** ($\max |u| = 2.2036$ vs. reference 2.2133).
- Formula B survives, but overshoots by 24.59% ($\max |u| = 2.7576$) due to its weaker $O(\Delta t^4)$ dissipation.

6 Target Application Domains and Conclusion

The Pratyaksh framework does not violate Dahlquist's barriers or seek to replace implicit solvers for stiff ODEs when variable step-size control is available. Instead, it establishes an explicit, matrix-free safeguard engineered for computational environments where step-rejection, Jacobian evaluation, or matrix inversions are prohibited. Five primary domains benefit immediately from this architecture:

1. **Real-Time Game Physics and Robotics (Unreal, PhysX, MuJoCo, Isaac Sim):** In hard real-time simulations running at 60 Hz or 120 Hz (16.6 ms or 8.3 ms per frame), adaptive step-reduction drops frame rates catastrophically, while implicit Newton-Raphson solvers have unpredictable convergence loops. The Pratyaksh framework executes in deterministic per-step FLOPs, converting unrecoverable NaN ragdoll explosions and joint divergence into detectable, bounded throttled states during sudden contact impulses.
2. **Massively Parallel GPU Machine Learning (PyTorch, JAX, Neural ODEs):** On GPU clusters running 10^5 parallel environments for reinforcement learning or continuous-depth neural networks, a single thread encountering stiffness stalls the entire warp if adaptive stepping is attempted. The Pratyaksh framework provides uniform, branchless execution engineered to eliminate warp divergence, serving as a promising architecture for stabilizing continuous-depth training loops without thread divergence.
3. **Hyperbolic Shockwaves and Aerodynamics (Fluids, Astrophysics, Plasma):** In compressible fluid dynamics and shock tube simulations where standard RK4 generates spurious Gibbs oscillations, Formula A provides built-in TVD shock capturing (0 total variation increases) without the overhead of non-linear Riemann flux limiters.
4. **Embedded Autonomous Systems and Flight Controllers (Drones, FPGAs, DSPs):** Low-power microcontrollers onboard autonomous aerial vehicles cannot allocate memory for $N \times N$ Jacobian matrices. The Pratyaksh integrator delivers bounded explicit time-stepping with zero matrix memory overhead, cushioning unexpected aerodynamic turbulence.

5. **High-Dimensional Molecular and Coarse-Grained Dynamics:** In large atomic systems ($N > 10^5$) where fast covalent bond vibrations trigger explicit stability limits, implicit solvers suffer from memory exhaustion ($O(N^2)$ storage). The Pratyaksh framework offers a matrix-free pathway to step through stiff transient modes without $O(N^2)$ memory overhead, representing an attractive candidate for future large-scale macromolecular simulation.

The reference single-header C++20 implementation, benchmark suites, and verification tests are preserved at: <https://github.com/Pratyaksh3142/The-Pratyaksh-Equation>.

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